

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**

**ASSESSMENT TOOL 3 – Technical Presentation**

|                         |                                                                                                                      |                      |                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| <b>Course Code:</b>     | CSA0305                                                                                                              | <b>Course Title:</b> | Data Structures                            |
| <b>CO Assessed:</b>     | CO1 – Imitation (BL3, BL4)                                                                                           | <b>Assessment:</b>   | Assessment Tool 3 – Technical Presentation |
| <b>Weightage:</b>       | 35% of CIA                                                                                                           | <b>Total Marks:</b>  | 100                                        |
| <b>CO1 Statement:</b>   | <i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i> |                      |                                            |
| <b>Student Name:</b>    | S.Roshni                                                                                                             | <b>Reg. No.:</b>     | 192525185                                  |
| <b>Section / Batch:</b> | AI&ML                                                                                                                | <b>Date:</b>         | 15/07/2026                                 |

**Code of Conduct**

*I S.Roshni(192525185) \_\_\_\_\_ (Name / Reg No)  
certify that this submission is my original work and that I have adhered to the guidelines specified for this  
assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S.Roshni\_\_\_\_\_

**Instructions**

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

| Section / Topic |  | Focus                  | Q Nos     | Marks |
|-----------------|--|------------------------|-----------|-------|
| Data Structure  |  | Real-Time Applications | 1         | 100   |
| GRAND TOTAL:    |  |                        | 100 Marks |       |

## Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

### Rubrics for Calculation

| Criteria              | Weightage | Level 4 – Exemplary                                                                          | Level 3 – Proficient                             | Level 2 – Developing                               | Level 1 – Beginning                         |
|-----------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|---------------------------------------------|
| Concept Understanding | 20        | Demonstrates deep and accurate understanding of data structure with insights and connections | Shows clear understanding with minor gaps        | Basic understanding; some misconceptions present   | Limited or incorrect understanding          |
| Content Organization  | 30        | Logical, well-structured, smooth flow; strong introduction and conclusion                    | Mostly organized with clear flow                 | Some organization but lacks clarity or transitions | Disorganized, difficult to follow           |
| Technical Depth       | 20        | Includes algorithms, complexity analysis, and advanced insights                              | Includes algorithm and basic complexity analysis | Limited technical details; missing depth           | Minimal or no technical content             |
| PowerPoint Slides     | 20        | Excellent design, clear visuals, enhances understanding                                      | Good slides with minor issues                    | Basic slides; somewhat cluttered or unclear        | Poor slides or no visual support            |
| Communication Skills  | 10        | Clear, confident, engaging delivery; excellent articulation                                  | Clear delivery with minor hesitation             | Some hesitation; unclear in parts                  | Poor communication; difficult to understand |



| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



# WEB BROWSER BACKWARD NAVIGATION

*Using Stack (Abstract Data Type)*



**Student Name:** S. Roshni

**Register Number:** 192525185

**Course Name:** Data Structures

Course Code: CSA0305

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TOP (Back removes this first)

Gmail

YouTube

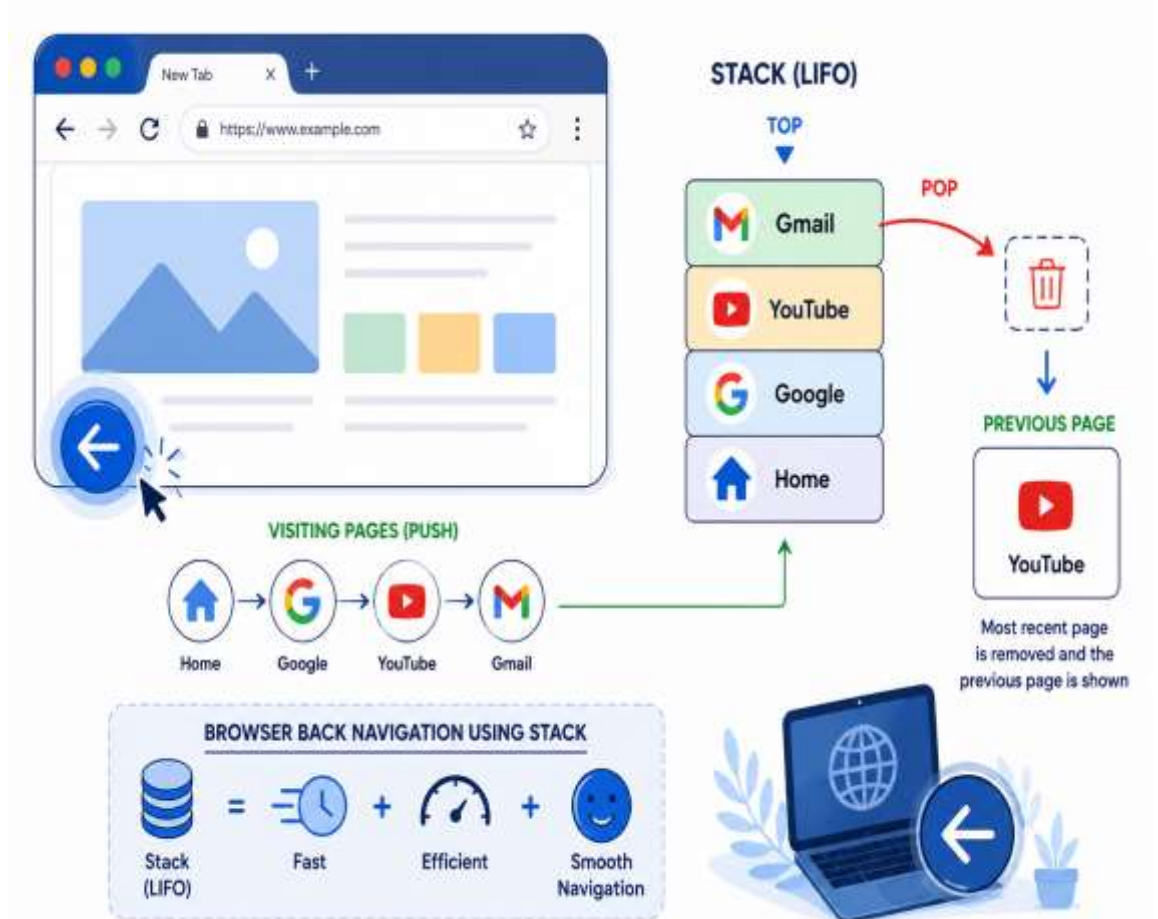
Google

Home



# Abstract

A web browser uses the Stack Abstract Data Type (ADT) to manage backward navigation efficiently. Each visited webpage is pushed onto the stack, and pressing the Back button pops the most recently visited page, following the Last In, First Out (LIFO) principle. This approach provides constant-time navigation ( $O(1)$ ), improves browser responsiveness, reduces processing overhead, and offers a simple yet efficient solution widely used in modern web browsers.



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